Fresh qubits set the memory: A sharp reset–memory law and classical learning of noisy quantum reservoirs

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How long can a noisy quantum device remember a classical input? We consider reservoirs that receive one input per cycle through at most k freshly prepared qubits, apply an arbitrary fixed unitary, and are then depolarized qubit by qubit. Once the initial state has washed out, a written bit stays readable for $\exp[\Theta_q(k)]$ cycles and no longer, independently of the total number n of qubits. Equivalently, keeping an input readable for T cycles costs $\Theta_q(\log T)$ fresh qubits per cycle. The upper bound follows from a single norm that every noisy update contracts by the same factor. The matching lower bound is a reversible majority-refresh reservoir whose exact response also rules out every shorter-window predictor. The same norm covers Markovian noise around a depolarizing core. Unital noise, including correlated, coherent, and drifting errors, leaves the law unchanged, and the law stays sharp for every nontrivial unital qubit channel with a full-rank Choi matrix. Nonunital noise counts as fresh-state supply, and this accounting is tight: with depolarization plus amplitude damping, two fresh qubits per cycle already give a memory $\exp[\Theta(n)]$. Pure dephasing and temporally correlated noise show that the depolarizing core and Markovianity are both needed. Finally, the upper bound makes driven reservoirs classically learnable: box-constrained least squares on a known library of input products, whose size does not depend on n, learns from the original task labels with a sample cost nearly linear in the library size.

I. INTRODUCTION

Imagine a 100-qubit reservoir that receives one classical input per cycle. A fixed unitary mixes the qubits, and between inputs every qubit is exposed to noise. Write a single bit into the device, then send only routine inputs. How long can a measurement still tell whether the bit was 0 or 1?

A larger register looks like a longer memory. The device could hold a hundred copies of the bit, and a majority vote over noisy copies is far more reliable than any single copy. But noise erodes every copy in every cycle, so the vote stays reliable only while the copies are repaired. Repair removes entropy, and the entropy has to go somewhere. In a reservoir, the only steady supply of low-entropy state is the freshly prepared qubits that arrive with each input. The question is central to quantum reservoir computing, where the internal dynamics is fixed and only a classical readout is trained for a temporal task [1]: whatever the task needs from the past must survive the device's own noise.

We prove that the fresh-state supply sets the answer. Suppose at most k qubits are reset per cycle, an arbitrary fixed unitary acts, and every qubit is then depolarized independently with a fixed parameter $0 < q < 1$. Once the initial state has washed out,

$$\text{longest readable memory} = \exp[\Theta_q(k)] \text{ cycles}, \qquad (1)$$

irrespective of the total number n of qubits. Equivalently, sustaining a readable input for T cycles requires, and suitable reservoirs achieve it with, $\Theta_q(\log T)$ fresh qubits per cycle. The upper bound holds for arbitrary quantum dynamics. A reversible classical circuit, embedded in the same architecture, attains the exponential order, so the obstruction is genuinely about fresh-state resources, not an assumption that the dynamics is classical.

The law does not hinge on the fine print of depolarization. From the noise, the proof needs only one layer of independent depolarization per cycle. Any unital noise may surround it, including correlated, coherent, and drifting errors, and the law stays sharp for every nontrivial unital qubit channel with a full-rank Choi matrix. Nonunital noise enters as additional fresh-state supply, and this accounting is tight: with depolarization plus amplitude damping, two fresh qubits per cycle already give a memory $\exp[\Theta(n)]$ that grows with the register. Pure dephasing and temporally correlated noise show that the remaining assumptions, a depolarizing layer and Markovianity, cannot be dropped. Together, these results add a reset axis to the classification of noisy computation without fresh ancillas by Ben-Or, Gottesman, and Hassidim [2].

Two different clocks are at work. An arbitrary initial state can contain many redundant copies of a bit, and this reserve takes $O_q(\log n)$ cycles to wash out. After the transient, adding qubits without adding fresh-state supply cannot make the worst-case memory diverge. Our matching construction writes its bit after the transient; it does not spend a large pure initialization. The bound also has a learning consequence: a reservoir that forgets is, seen from outside, a formula in its recent inputs, which can be learned from task labels without simulating the quantum state.

Register size delays forgetting; the supply of fresh qubits decides it.

The paper is organized as follows. Section II reduces the optimal memory to a one-line recurrence. Section III

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states the model and the sharp law, Theorem 1, whose two halves are proved in Secs. IV and V. Section VI turns the upper bound into a learning guarantee, Sec. VII extends the law beyond depolarizing noise, and Sec. VIII relates it to previous work.

II. A MEMORY IN ONE LINE

Start with the simplest memory: store a bit in one qubit as $|0\rangle$ or $|1\rangle$ and measure Z later. Depolarization, $\mathcal{D}_q(X) = qX + (1-q)\,\mathrm{Tr}(X)\,I/2$, keeps the qubit with weight q and replaces it by the completely mixed state with weight $1-q$. On a stored bit, it is a flip with probability $p = (1-q)/2$ in each cycle, so after t cycles the readout is $\pm q^t$. At $q = 0.9$, the signal falls below half its initial size after seven cycles.

A. Copies help only while they are repaired

Store the bit in r qubits instead, with r odd. A majority vote errs only if more than half of the copies are wrong. Left alone, however, every copy drifts toward a coin flip, and the vote follows: untouched copies stay decodable for only logarithmically many cycles in r (Sec. IV E and Appendix B).

The cure is to repair the copies in every cycle. Compute the majority of the r noisy copies, write it into r freshly prepared qubits, and discard the old ones. The errors leave with the discarded qubits, and each cycle starts again from r clean copies. The stored bit now changes only if more than half of the copies flip within one cycle, which happens with probability

$$p_r = \Pr\{\mathrm{Bin}(r,p) > r/2\}. \tag{2}$$

B. The memory is one number

Let $m_t = \mathbb{E}[(-1)^{B_t}]$ be the bias of the stored bit B_t after t repair cycles, so $m_0 = \pm 1$ at the write. A logical flip occurs with probability p_r in every cycle, whichever value is stored. Hence

$$m_t = \underbrace{(1-2p_r)}_{\lambda_r}\, m_{t-1}, \qquad f_t = \underbrace{q}_{\text{readout noise}}\, m_t. \tag{3}$$

The first equation is the entire memory. The second reads it through one qubit, which pays the factor q for its own latest round of noise. A bit written t cycles ago appears in the output as $\pm q\lambda_r^t$; for $r = 1$, $\lambda_1 = q$ and nothing is gained.

How close is λ_r to one? An exponential-moment estimate (Sec. V) gives

$$p_r \le (1-q^2)^{r/2} = e^{-\alpha_q r}, \qquad \alpha_q = -\tfrac{1}{2}\log(1-q^2). \tag{4}$$

The bound shrinks by the factor $\sqrt{1-q^2}$ per additional fresh copy, and the signal stays above half its initial size for at least $e^{\alpha_q r}/4$ cycles. *The memory grows exponentially with the number of fresh qubits per cycle.* Table I gives exact values at $q = 0.9$: seventeen fresh qubits per cycle keep a bit readable for almost two million cycles, and every two more multiply the lifetime by about six.

Repetition is not clever; the example matters because of the question it leaves open. Could a smarter reservoir, with entangling non-Clifford gates, many more qubits, and the same fresh supply, remember far longer, say polynomially in its total size? We prove that it cannot: no reservoir in this model beats the exponential order in k, and the majority reservoir attains it. *Exponentially long memory costs only linearly many fresh qubits per cycle, and no reservoir buys it more cheaply.*

III. MODEL AND MAIN RESULT

The general reservoir replaces the r copies by an n-qubit state ρ_t. At each cycle, a register A of k qubits is overwritten by a known state $\sigma_A(u_t)$ that encodes the classical input u_t. A fixed unitary U then acts on all n qubits. Finally, independent noise acts on every qubit. Writing these operations in the order in which they occur gives

$$\rho_t = \Phi_{u_t}(\rho_{t-1}), \qquad \Phi_u = \mathcal{N}_q \circ \mathcal{E}_u, \tag{5}$$

$$\mathcal{E}_u(X) = U\,[\sigma_A(u) \otimes \mathrm{Tr}_A\, X]\,U^\dagger, \tag{6}$$

$$\mathcal{N}_q = \mathcal{D}_q^{\otimes n}, \quad \mathcal{D}_q(X) = qX + \tfrac{1-q}{2}\,\mathrm{Tr}(X)\,I, \tag{7}$$

with $0 < q < 1$. The partial trace Tr_A discards whatever the reset register held, and the tensor product inserts its fresh replacement. The output is a bounded expectation value $f_t = \mathrm{Tr}(O\rho_t)$ with $O = O^\dagger$ and $\|O\|_\infty \le 1$, not a persistent external record of measurement outcomes. The unitary U is the same in every cycle but otherwise unrestricted: it can be nonlocal, deep, and non-Clifford. All times count the ideal-unitary/noise cycles in Eq. (5); Section VII generalizes the noise.

Where the fresh qubits enter. The upper bound uses a single property of the update. It allows any completely positive, trace-preserving (CPTP) maps satisfying, for some $J \ge 1$,

$$\mathcal{E}_u(I) \le J\,I \quad \text{uniformly in } u. \tag{8}$$

For Eq. (6), $\mathcal{E}_u(I) = 2^k U[\sigma_A(u)\otimes I]U^\dagger \le 2^k I$, so $J = 2^k$ suffices. Thus J bounds the purity that one update can inject: it lowers the min-entropy of the completely mixed state by at most $\log_2 J = k$ bits. Here k is a per-cycle budget, not the total number of reset operations over a run. Section VII measures nonunital noise in exactly the same way.

TABLE I. Exact readable lifetime of the majority-refresh reservoir of Sec. V at $q = 0.9$, so $p = 0.05$, with $k = r + 2$ fresh qubits per cycle. The lifetime T_r is the largest T with $\lambda_r^T \geq 1/2$, so the readout gap between a written 0 and a written 1 stays at least q. The guarantee $\lfloor e^{\alpha_q r}/4 \rfloor$ from Eq. (4) has the right exponential rate but a loose prefactor.

Copies r	1	3	5	7	9	11	13	15
Fresh qubits per cycle k	3	5	7	9	11	13	15	17
Logical error per cycle p_r	5.0×10^{-2}	7.3×10^{-3}	1.2×10^{-3}	1.9×10^{-4}	3.3×10^{-5}	5.8×10^{-6}	1.0×10^{-6}	1.8×10^{-7}
Lifetime T_r (cycles)	6	47	298	1 790	10 431	59 739	337 940	1 894 286
Guarantee $\lfloor e^{\alpha_q r}/4 \rfloor$	0	3	15	83	440	2 316	12 191	64 163

A. Memory measured by the window it needs

A left-infinite input history $h = (\ldots, u_{-1}, u_0)$ has a unique driven state (Sec. IV E); let $f(h)$ denote its output. To state sharpness without committing to any classical architecture, let $W_f(\varepsilon)$ be the smallest $L \in \mathbb{N}_0$ for which some function g of the last L inputs satisfies

$$\sup_h \big| f(h) - g(u_0, \ldots, u_{-L+1}) \big| \leq \varepsilon. \tag{9}$$

The function g is unrestricted. Inputs older than $W_f(\varepsilon)$ can be forgotten at cost ε; newer ones cannot. Let $W_{q,k}(\varepsilon)$ be the supremum of $W_f(\varepsilon)$ over all register sizes, encodings, fixed unitaries, and bounded readouts in this model.

Theorem 1 (Sharp reset-width scaling)**.** *Fix $0 < q < 1$, and define*

$$\alpha_q = -\tfrac{1}{2}\log(1-q^2), \qquad \beta_q = \frac{q(1+q^2)}{(1-q)(1-q^2)}. \tag{10}$$

There is an explicit $A_q > 0$, depending only on q, such that for all sufficiently large k and $0 < \varepsilon < q/2$,

$$\begin{aligned} \tfrac{1}{4}\, e^{\alpha_q (k-3)} \log \frac{q}{\varepsilon} &\leq W_{q,k}(\varepsilon) \\ &\leq \left\lceil A_q\, 2^{\beta_q k} \log \frac{2}{\varepsilon} \right\rceil. \end{aligned} \tag{11}$$

The lower bound is attained by a reservoir with $n \leq 2k$ and a single-qubit Z readout. The upper bound holds for arbitrary bounded readouts and arbitrary register size. For arbitrary finite initializations, the upper window bound needs only an additional $O_q(\log n)$ initialization time.

Both sides of Eq. (11) grow exponentially in k and neither depends on n, so for fixed q and accuracy $W_{q,k} = \exp[\Theta_q(k)]$, the law (1). The exponents α_q and $\beta_q \log 2$ differ: the theorem is sharp in its order in k, not an equality of constants, and it limits finite-window replacement, not every classical algorithm (Sec. V). Section IV proves the upper bound; Sec. V builds the reservoir that attains the lower one.

IV. WHY EVERY RESERVOIR FORGETS

In the example, forgetting was visible: one number shrank by λ_r per cycle. A general reservoir can hide information much better: it can spread a bit over many qubits, entangle it, and keep repairing it with its fresh qubits. A proof of forgetting must defeat every such strategy at once, without looking at what U does. The proof therefore tracks two properties. *Distinguishability* measures how well some measurement can tell two histories apart; *concentration* measures how far a state is from completely mixed. Noise lowers concentration, and once concentration is bounded, noise also lowers distinguishability. Together they give one distance that contracts under every update.

Write $d = 2^n$ and use the normalized Schatten quantities $\tau(X) = \mathrm{Tr}(X)/d$ and $|X|_p = (\tau|X|^p)^{1/p}$. In particular, $|d(\rho - \sigma)|_1 = \|\rho - \sigma\|_1$ is the trace distance, which bounds every measurement, while $|d\rho|_2 = \sqrt{d\,\mathrm{Tr}\,\rho^2}$ equals 1 for the completely mixed state and $2^{n/2}$ for a pure state. Every allowed update satisfies, for Hermitian X,

$$|\mathcal{E}X|_1 \leq |X|_1, \qquad |\mathcal{E}X|_2 \leq \sqrt{J}\,|X|_2. \tag{12}$$

The first is trace-norm contractivity. For the second, the Schwarz inequality for the subunital completely positive map $\mathcal{E}/J$ gives $(\mathcal{E}X)^\dagger(\mathcal{E}X) \leq J\,\mathcal{E}(X^\dagger X)$; take traces.

A. The two noise inequalities

Two established inequalities are the only specialized tools in the proof. Both concern $\mathcal{N}_q$, and all normalized quantities in them use the same τ. King's depolarizing hypercontractivity theorem [3, Corollary 2] says that noise spreads operators out:

$$|\mathcal{N}_q Y|_2 \leq |Y|_{1+q^2}. \tag{13}$$

Indeed, its condition $q \leq \sqrt{(p-1)/(r-1)}$ holds with equality for $p = 1 + q^2$ and $r = 2$. The two-point reverse inequality of Cubitt *et al.* [4, Corollary 14] says that noise forces positive operators to overlap:

$$\tau[F\,\mathcal{N}_q(G)] \geq |F|_{\bar{p}}\,|G|_{\bar{p}}, \qquad F, G \geq 0, \quad \bar{p} = 1 - q. \tag{14}$$

Here the required condition $q \leq \sqrt{(1-\bar{p})(1-\bar{p})}$ is again an equality. For $\bar{p} < 1$, $|\cdot|_{\bar{p}}$ is not a norm; on positive G with $\tau G = 1$ it is at most one, and small only when G is concentrated.

B. Noise caps concentration

Lemma 2 (Concentration cannot remain arbitrarily large)**.** *For any CPTP $\mathcal{E}$ satisfying $\mathcal{E}(I) \le JI$ and Hermitian X, put $\theta = 2q^2/(1+q^2)$ and $C = J^{\theta/2}$. Then*

$$|\mathcal{N}_q\mathcal{E}(X)|_2 \le C\,|X|_1^{1-\theta}|X|_2^{\theta}. \tag{15}$$

Proof. Since $1/(1+q^2) = 1-\theta/2$, interpolation between the 1- and 2-norms gives $|Y|_{1+q^2} \le |Y|_1^{1-\theta}|Y|_2^{\theta}$. Apply it to $Y = \mathcal{E}X$, then use Eqs. (13) and (12). □

The update can raise the 2-norm by at most $\sqrt{J}$; noise then pulls it down toward the 1-norm. For a density matrix, $X = d\rho$ has $|X|_1 = 1$, so the concentration $P_t = |d\rho_t|_2$ obeys

$$P_t \le C\,P_{t-1}^{\theta}. \tag{16}$$

In logarithms, $\log P_t \le \log C + \theta \log P_{t-1}$: a fading-memory rule for purity. Each cycle multiplies the remembered log-purity by $\theta < 1$, while the fresh qubits add at most $\log C = \frac{\theta}{2}\log J$.

C. Bounded concentration forces forgetting

An upper bound on purity or entropy deficit alone cannot imply forgetting. The two states $\rho_b = |b\rangle\langle b| \otimes I/2^{n-1}$, $b \in \{0,1\}$, have $|d\rho_b|_2 = \sqrt{2}$ and entropy deficit one bit, yet $\|\rho_0 - \rho_1\|_1 = 2$: a bounded amount of information can still encode a bit perfectly. The missing step is a uniform loss of distinguishability under the next noisy update, and the reverse inequality supplies it.

Lemma 3 (Bounded concentration forces distinguishability loss)**.** *Let $Y \neq 0$ be traceless and Hermitian. Set $s = |Y|_1$, $t = |Y|_2$, $a = 2q/(1-q)$, and $c_q = 2^{-1/(1-q)}$. Then*

$$|\mathcal{N}_q Y|_1 \le s\Big[1 - c_q\Big(\frac{s}{2t}\Big)^a\Big]. \tag{17}$$

Read Eq. (17) as a trade. When s/t is not small, the difference Y is spread out, and one layer of noise must destroy a fixed fraction of its distinguishability. The proof (Appendix A) splits Y into its positive and negative parts and uses Eq. (14) to show that every measurement confuses them with probability bounded below.

D. One distance contracts under every update

Theorem 4 (Uniform contraction under bounded fresh-state supply)**.** *Using the constants above, define*

$$R = (2C)^{1/(1-\theta)}, \qquad \delta = c_q(4\sqrt{J}R)^{-a},$$
$$V(X) = \max\{|X|_1,\, |X|_2/R\}. \tag{18}$$

Then $0 < \delta \le 1/2$ and, for every traceless Hermitian X and every allowed update,

$$V(\mathcal{N}_q\mathcal{E}X) \le (1-\delta)\,V(X). \tag{19}$$

The norm and contraction rate are independent of n, the input sequence, and the choice of gates.

Proof. By homogeneity assume $V(X) = 1$. Lemma 2 gives $|\mathcal{N}_q\mathcal{E}X|_2/R \le CR^{\theta-1} = 1/2$. Put $s = |\mathcal{E}X|_1 \le 1$ and $t = |\mathcal{E}X|_2 \le \sqrt{J}R$, by Eq. (12). If $s \le 1/2$, then $|\mathcal{N}_q\mathcal{E}X|_1 \le 1/2$. If $s > 1/2$, Lemma 3 gives

$$\begin{aligned}|\mathcal{N}_q\mathcal{E}X|_1 &\le s\big[1 - c_q(s/2t)^a\big] \\ &\le 1 - c_q(4\sqrt{J}R)^{-a} = 1-\delta.\end{aligned} \tag{20}$$

Both components of the maximum are at most $1-\delta$. Finally, $c_q \le 1/2$ and $4\sqrt{J}R > 1$ give $0 < \delta \le 1/2$. □

This proof needs no strong-noise condition such as $q\sqrt{J} < 1$, and it makes the resource dependence explicit. Since $R = 2^{1/(1-\theta)}J^{\theta/[2(1-\theta)]}$,

$$\delta^{-1} = A_q J^{\beta_q}, \qquad \beta_q = \frac{a}{2(1-\theta)},$$
$$A_q = 2^{1/(1-q)+a[2+1/(1-\theta)]}. \tag{21}$$

For a k-qubit reset register this is exactly $\delta^{-1} = A_q 2^{\beta_q k}$, with β_q as in Eq. (10): each fresh qubit can stretch the memory by at most the constant factor 2^{β_q}. The constants are explicit but not small. At $q = 1/2$ and $k = 1$, $\delta^{-1} = 2^{11}$; at $q = 0.9$, $\delta^{-1} \approx 2^{217.5+85.7k}$, far above the exact lifetimes in Table I. The theorem fixes the order in k, not the constant.

E. Washout first, then driven forgetting

Iterating Eq. (16) from $P_0 \le 2^{n/2}$ gives

$$P_t \le P_*\,2^{n\theta^t/2}, \qquad P_* = C^{\frac{1}{1-\theta}} = J^{\frac{q^2}{1-q^2}}. \tag{22}$$

The initial log-purity is proportional to n, but its contribution is multiplied by θ in every cycle. After $t_0 = \lceil \log n/(-\log\theta)\rceil$ cycles, $P_t \le \sqrt{2}P_*$, and this region is invariant under every allowed update. Since $R \ge 2P_*$, the difference of any two states in this region has $V(d(\rho-\sigma)) \le 2$. Repeated application of Eq. (19) proves

$$\|\Phi_{u_L}\cdots\Phi_{u_1}(\rho-\sigma)\|_1 \le 2(1-\delta)^{L-t_0}, \qquad L \ge t_0. \tag{23}$$

For a left-infinite input history, start at progressively earlier times from arbitrary states. By Eq. (23), the resulting states form a Cauchy sequence, independent of initialization. The limit is the unique driven state, and taking the starting time to minus infinity in Eq. (22) gives $P \le P_*$. Consequently, two driven histories that agree on their last L inputs satisfy

$$\|\rho(h) - \rho(h')\|_1 \le 2(1-\delta)^L \le 2e^{-\delta L}. \tag{24}$$

The same estimate compares a driven history to a window initialized at I/d, since both boundary states have V-difference at most two. Thus

$$L_\varepsilon = \lceil \delta^{-1} \log(2/\varepsilon) \rceil \tag{25}$$

recent inputs suffice for uniform output error ε. In a finite run, this window suffices at every time $t \geq t_0 + L_\varepsilon$. Equations (21) and (25) prove the upper half of Theorem 1.

The washout term is real. With $U = I$, m untouched copies of a bit give a total Z signal $\pm m q^L$ against fluctuations of order $\sqrt{m}$, so they stay decodable for $\Omega_q(\log n)$ cycles (Appendix B). A 1000-qubit reservoir thus takes longer than a 100-qubit one to forget its initialization (t_0 grows from 42 to 63 cycles at $q = 0.9$), but after washout both obey the same worst-case memory bound. *Forgetting is a property of each noisy update, not of the data, the gates, or the register size.*

V. A MEMORY THAT MEETS THE BOUND

We now build the memory of Sec. II from one fixed unitary and count every fresh qubit. The construction is classical logic implemented as a quantum unitary; it uses no measurements, no noiseless external record, and no pure data initialization. A single classical input writes the bit; every subsequent input is the same hold symbol.

A. One fixed unitary, with every fresh qubit counted

Take an odd $r \geq 1$. The data register D has r qubits. The reset register contains r work qubits W and two program qubits (c, b), so $k = r + 2$ and $n = 2r + 2$. Prepare the input register in one of three product states,

$$\text{hold}: \ |0^r\rangle|0,0\rangle, \qquad \text{write}(b): \ |0^r\rangle|1,b\rangle, \tag{26}$$

with $b \in \{0,1\}$. For a data string x, define $F_{c,b}(x) = \mathrm{maj}(x)$ when $c = 0$, and $F_{c,b}(x) = b$ when $c = 1$. The single fixed unitary is the basis permutation

$$U: \ |a,c,b,x\rangle \longmapsto |x,c,b,a \oplus F_{c,b}(x) 1_r\rangle. \tag{27}$$

It is reversible: the output reveals x, c, b, and therefore also a. Operationally, U computes the selected bit into the fresh work block and swaps the work and data blocks. The old data become garbage in W and are discarded by the next reset.

On a hold input, the new data are a repetition of the old majority. On a write input, they are $|b\rangle^{\otimes r}$, regardless of the old data state. All n qubits are then depolarized, including the program and garbage registers. Those two registers are reset before their next use, so their post-cycle noise supplies no hidden resource. Equation (27) specifies the unitary on the entire Hilbert space; no local-depth implementation is assumed.

B. The exact response is the one-line memory

All states stay diagonal in the computational basis, where depolarization flips each bit independently with probability $p = (1-q)/2$. Let B_t be the bit re-encoded before the noise at cycle t, with $B_0 = b$ at the write cycle. Conditioned on B_t, the post-cycle data consist of r independent noisy copies of that bit. Each hold cycle therefore flips the logical bit with conditional probability p_r of Eq. (2), the same for either value. The physical noise is independent between cycles. Hence $\mathbb{E}[(-1)^{B_{t+1}}] = \lambda_r \mathbb{E}[(-1)^{B_t}]$ with $\lambda_r = 1 - 2p_r \in (0,1)$, and measuring Z on the first data qubit after the cycle's noise yields the exact expectation

$$f_b(t) = (-1)^b \, q \, \lambda_r^t. \tag{28}$$

This is Eq. (3), now derived for the physical unitary. The factor q is local readout attenuation; the exponentially small logical error controls the memory time.

The bound (4) is an exponential-moment calculation. If $X_i \in \{-1, 1\}$ are independent with mean q, then for any $\eta > 0$,

$$p_r \leq \mathbb{E}\, e^{-\eta \sum_i X_i} = \left[\tfrac{1+q}{2} e^{-\eta} + \tfrac{1-q}{2} e^{\eta} \right]^r. \tag{29}$$

Choosing $e^{2\eta} = (1+q)/(1-q)$ gives $p_r \leq (1-q^2)^{r/2} = e^{-\alpha_q r}$. In particular, $\lambda_r^t \geq 1 - 2tp_r \geq 1/2$ whenever $t \leq e^{\alpha_q r}/4$, so the one-qubit signal has magnitude at least $q/2$ throughout that interval. The exponent is the right one: α_q is the exact large-r decay rate of the majority error, and $p_{r+2}/p_r \to 1 - q^2$.

A terminal majority measurement provides another operational readout: it decodes the original bit after t hold cycles with probability $(1 + \lambda_r^{t+1})/2$. It is only a possible final measurement; the memory dynamics uses the reversible circuit (27), not external classical feedback.

C. No shorter-window predictor can avoid the bit

Consider two driven histories that are identical before time $-L$, carry write(0) or write(1) at time $-L$, and then receive L hold inputs. A function of the last L inputs sees the same suffix in both cases, but the outputs at time zero are $\pm q \lambda_r^L$. Consequently, every such function has worst-case error at least $q\lambda_r^L$, and

$$W_f(\varepsilon) \geq \frac{\log(q/\varepsilon)}{-\log \lambda_r}, \qquad 0 < \varepsilon < q. \tag{30}$$

When $p_r \leq 1/4$, the elementary inequality $-\log(1 - 2p_r) \leq 4p_r$ and Eq. (4) give

$$W_f(\varepsilon) \geq \tfrac{1}{4} e^{\alpha_q r} \log(q/\varepsilon). \tag{31}$$

For a general budget k, choose the largest odd $r \leq k-2$ and pad the reset register with unused fresh qubits. Then $r \geq k - 3$ and $n = k + r \leq 2k$. The condition $k \geq$

$3 + \lceil \log 4/\alpha_q \rceil$ suffices for $p_r \le 1/4$. This proves the lower half of Theorem 1.

The lower bound cannot be attributed to initialization. The two histories can share an arbitrarily long common past, and the write sets the data regardless of their previous state. They are therefore valid histories of the unique driven reservoir.

Corollary 5 (Fresh-qubit cost of a readable input)**.** *Fix q. Require that two driven histories differing in a written bit and subsequently receiving the same hold inputs retain a readout gap of at least q for T cycles. The minimum per-cycle reset budget, optimized over register size and allowed reservoirs, is $\Theta_q(\log T)$ as $T \to \infty$.*

Proof. Equation (24) implies $q \le 2e^{-\delta T}$, hence $T \le A_q 2^{\beta_q k} \log(2/q)$ and $k = \Omega_q(\log T)$. For the construction, Eq. (28) has gap at least q at all times through T if $T \le e^{\alpha_q r}/4$. An odd $r \ge \log(4T)/\alpha_q$ and $k = r + 2$ give $k = O_q(\log T)$. □

Long memory is not classical hardness. This example is classically easy: maintain one scalar m_t, set $m_t = (-1)^b$ on a write and $m_t = \lambda_r m_{t-1}$ on a hold, and output $q m_t$. The window bound (30) therefore limits finite-window replacement, not classical computation in general. *Long memory and classical hardness are different resources.*

VI. LEARNING THE ANSWER FROM TASK LABELS

The upper bound says more than that a reservoir forgets. It says that, up to ε, the output is a function of the last L_ε inputs. We now make that function explicit and learnable. The idea fits in one line.

A. A recurrence that is a formula

Consider the scalar reservoir

$$y_t = \underbrace{\tfrac{1}{2}\sin u_t}_{\text{new input}} + \underbrace{\tfrac{1}{2}\cos u_t\, y_{t-1}}_{\text{retained memory}}, \tag{32}$$

with $|y_{t-1}| \le 1$. Substituting yesterday's rule into today's, and repeating, opens up the memory:

$$\begin{aligned} y_t = {} & \tfrac{1}{2}\sin u_t + \tfrac{1}{4}\cos u_t \sin u_{t-1} \\ & + \tfrac{1}{8}\cos u_t \cos u_{t-1} \sin u_{t-2} + \cdots . \end{aligned} \tag{33}$$

Each term is the contribution of one past input, carried through the updates since. Truncating after L terms leaves an error of at most 2^{-L} when $|y_{t-L}| \le 1$; at accuracy 0.01, seven terms suffice, since $2^{-7} \approx 0.0078$. A learner who did not know the weights $\frac{1}{2}, \frac{1}{4}, \ldots$ would compute seven columns of products and fit one weight per column. The products are nonlinear in the inputs; fitting their weights is linear regression. In the quantum reservoir, the contraction theorem supplies the window, and products of known input coordinates supply the expansion. The quantum dynamics determines the coefficients, which ordinary regression learns without simulating the state.

B. A known library with bounded coefficients

For this section, use the erase-and-write architecture (6) and assume the 4^k Pauli coordinates of the known input state can be evaluated classically to arbitrary polynomial precision. No classical simulation of U is assumed. Let $m = 2^k$ and let P_a, $0 \le a < 4^k$, be the k-qubit Pauli matrices. Write

$$\begin{aligned} \phi_a(u) &= \mathrm{Tr}[P_a \sigma_A(u)], \\ \sigma_A(u) &= \frac{1}{m}\sum_a \phi_a(u) P_a, \end{aligned} \tag{34}$$

so that $|\phi_a(u)| \le 1$. The update is linear in these known coordinates:

$$\begin{aligned} \Phi_u &= \sum_a \phi_a(u)\, \mathcal{A}_a, \\ \mathcal{A}_a(X) &= \mathcal{N}_q\big(U[P_a/m \otimes \mathrm{Tr}_A\, X]U^\dagger\big). \end{aligned} \tag{35}$$

Every $\mathcal{A}_a$ contracts the ordinary trace norm on Hermitian matrices: partial trace is contractive on that space, $\|P_a/m\|_1 = 1$, and the remaining maps are quantum channels. Starting a window of length L from I/d therefore gives the exact expansion

$$\begin{aligned} & f_L(u_0, \ldots, u_{-L+1}) \\ & \quad = \sum_{a_0,\ldots,a_{L-1}} c_{a_0 \cdots a_{L-1}} \prod_{j=0}^{L-1} \phi_{a_j}(u_{-j}), \end{aligned} \tag{36}$$

with coefficients

$$c_{a_0 \cdots a_{L-1}} = \mathrm{Tr}\big[O\, \mathcal{A}_{a_0} \cdots \mathcal{A}_{a_{L-1}}(I/d)\big], \tag{37}$$

which satisfy $|c_{a_0 \cdots a_{L-1}}| \le 1$. By Sec. IV E, $\|f - f_L\|_\infty \le \varepsilon$ for $L = L_\varepsilon$. The library has size

$$K = (4^k)^L \le 4^k (2/\varepsilon)^{k \log 4 \cdot A_q 2^{\beta_q k}}. \tag{38}$$

For fixed q, k, this is polynomial in $1/\varepsilon$ and independent of n in driven operation. Even a window long enough to cover an arbitrary initialization, $L = t_0 + L_\varepsilon$, gives $K = \mathrm{poly}_{q,k}(n, 1/\varepsilon)$.

Concretely, one fresh qubit prepared by an $R_y(u)$ rotation has $\sigma(u) = \frac{1}{2}[I + \sin(u)X + \cos(u)Z]$, so only 1, $\sin u$, and $\cos u$ are needed and $K \le 3^L$. Two steps give nine products, of which the scalar example (33) uses two; a quantum reservoir can use all of them, with different weights. Such temporal Fourier expansions are established [5]; what is new is the physical window, its sharp reset-width scaling, coefficient control, and learning. *The quantum dynamics chooses the weights; the encoding supplies the building blocks.*

C. Learning from the original labels

Let $(H_i, Y_i)_{i=1}^N$ be independent, identically distributed classical task examples, with $|Y_i| \le 1$; for instance, H_i is a history of sensor readings and Y_i the next reading. Let $\psi(H_i) \in [-1,1]^K$ be the products in Eq. (36), evaluated on the last L inputs. With the empirical objective

$$F_N(c) = \frac{1}{N}\sum_{i=1}^{N}\big(c^\top \psi(H_i) - Y_i\big)^2, \tag{39}$$

fit

$$\hat{c} \approx \operatorname*{arg\,min}_{c\in[-1,1]^K} F_N(c), \qquad \hat{f} = \mathrm{clip}_{[-1,1]}(\hat{c}^\top \psi). \tag{40}$$

Training is a convex quadratic problem; clipping is applied only after training. The coefficient bound $|c_{a_0\cdots a_{L-1}}| \le 1$ supplies the box. *The labels are the original task labels, not quantum-generated calibration targets.*

Theorem 6 (Classical task learning with an explicit sample bound)**.** *Suppose Eq. (40) is solved to empirical objective error $\nu \ge 0$ and $L = L_\varepsilon$. Let $\mathcal{R}(g) = \mathbb{E}[(g(H) - Y)^2]$. With probability at least $1-\zeta$,*

$$\mathcal{R}(\hat{f}) \le \mathcal{R}(f) + 4\varepsilon + \nu + \frac{16}{N} + \sqrt{\frac{32}{N}\Big(K\log(2KN+1) + \log\frac{2}{\zeta}\Big)}. \tag{41}$$

Thus excess risk Δ is achievable with $\varepsilon, \nu = O(\Delta)$ and $N = \tilde{O}((K + \log(1/\zeta))/\Delta^2)$, with polynomial classical computation.

Each term has a job: 4ε pays for the discarded past, ν for approximate optimization, and the last two terms for finite data. The proof (Appendix C) compares $\hat{f}$ with the physical window output f_L, which lies in the same box-constrained class, and controls the generalization gap with a finite grid used only in the analysis; the resulting nearly linear, rather than quadratic, dependence on K is a standard statistical improvement, not new learning theory. Appendix C also gives a polynomial-time solver and shows that finite precision needs no quantum simulation.

VII. BEYOND DEPOLARIZING NOISE

Real devices do not depolarize every qubit in exactly the same way: their errors are anisotropic, partly coherent, correlated, and they drift. Does the law survive? The proof of Sec. IV used only two properties of the noise: a layer of independent depolarization in each cycle, and bounded purity injected before it. Both survive a great deal of additional noise.

A. Noise around a depolarizing core

Replace the noise layer by

$$\mathcal{N}_t = \mathcal{R}_t \circ \mathcal{D}_q^{\otimes n} \circ \mathcal{S}_t, \tag{42}$$

where $\mathcal{R}_t$ and $\mathcal{S}_t$ are CPTP maps on all n qubits. They may be correlated across qubits, contain coherent errors, and change from cycle to cycle. Their nonunitality is measured exactly like that of the reset in Eq. (8): $\mathcal{R}_t(I) \le J_{\mathcal{R}} I$ and $\mathcal{S}_t(I) \le J_{\mathcal{S}} I$. Unital noise, such as dephasing, Pauli errors, coherent over-rotations, or any mixture of unitaries, has $J_{\mathcal{R}} = J_{\mathcal{S}} = 1$. Because unitaries are unital, the depolarizing layer may also sit anywhere after the reset, between noisy unital gates.

Theorem 7 (Noise around a depolarizing core)**.** *Replace $\mathcal{N}_q$ by $\mathcal{N}_t$ from Eq. (42), with $\mathcal{R}_t$ independent of the inputs. Then Eqs. (23) and (24), the upper half of Theorem 1, and the bound $k = \Omega_q(\log T)$ of Corollary 5 hold at every time with k replaced by $k + \log_2(J_{\mathcal{R}} J_{\mathcal{S}})$; apart from the washout time, the register size enters only through $J_{\mathcal{R}} J_{\mathcal{S}}$. If $\mathcal{R}_t$ and $\mathcal{S}_t$ are unital, they may also depend on the inputs, Eq. (19) holds with the same δ, and Theorem 1 holds verbatim when the supremum defining $W_{q,k}$ also runs over all such maps.*

Proof. Start with unital $\mathcal{R}_t$ and $\mathcal{S}_t$. The map $\mathcal{S}_t \circ \mathcal{E}_u$ is CPTP, and $\mathcal{S}_t(\mathcal{E}_u(I)) \le J\,\mathcal{S}_t(I) = JI$ because $\mathcal{S}_t$ is positive and unital; so it is an allowed update in Theorem 4. A unital CPTP map $\mathcal{R}$ cannot increase V: it contracts the trace norm, and the Schwarz inequality $\mathcal{R}(Y)^\dagger \mathcal{R}(Y) \le \mathcal{R}(Y^\dagger Y)$, traced, gives $|\mathcal{R}Y|_2 \le |Y|_2$. Hence $V(\mathcal{N}_t\mathcal{E}_u X) \le V(\mathcal{N}_q\mathcal{S}_t\mathcal{E}_u X) \le (1-\delta)V(X)$, and the same two facts give Eq. (16). The lower half of Theorem 1 needs only $\mathcal{R}_t = \mathcal{S}_t = \mathrm{id}$.

In general, regroup the cycles. The state after cycle t is $\rho_t = \mathcal{R}_t(\tilde{\rho}_t)$, where $\tilde{\rho}_t = \Psi_t(\tilde{\rho}_{t-1})$ and $\Psi_s = \mathcal{N}_q \circ (\mathcal{S}_s \circ \mathcal{E}_{u_s} \circ \mathcal{R}_{s-1})$, with $\mathcal{R}_0 = \mathrm{id}$ at the start of a run. By positivity, $\mathcal{S}_s\mathcal{E}_{u_s}\mathcal{R}_{s-1}(I) \le J J_{\mathcal{R}} J_{\mathcal{S}}\, I$, so each Ψ_s is an allowed update for this larger J, and Sec. IV applies to $\tilde{\rho}_t$. Because $\mathcal{R}_{t-L}$ does not depend on the inputs, two histories that share their last L inputs share every Ψ_s in that window; the comparison window starts from $\mathcal{R}_{t-L}(I/d)$, which is I/d in the unital case. Finally, $f_t = \mathrm{Tr}[\mathcal{R}_t^*(O)\,\tilde{\rho}_t]$, and the dual $\mathcal{R}_t^*$ is unital and positive, so $\|\mathcal{R}_t^*(O)\|_\infty \le 1$. □

The bounds hold for every realization of random noise, hence also on average. The learning results of Sec. VI carry over when $\mathcal{R}_t$ and $\mathcal{S}_t$ are unital and either fixed, or random and independent between cycles and of the inputs: the maps $\mathcal{A}_a$ in Eq. (35) then absorb the averaged noise and remain trace-norm contractions.

B. Every nontrivial unital qubit channel

Where is the depolarizing core in a given noise model? For product noise $\mathcal{M}^{\otimes n}$, it can be read off the Choi matrix. Let $\mathcal{M}$ be a unital qubit channel, $J_{\mathcal{M}} = \sum_{ij} |i\rangle\langle j| \otimes$

$\mathcal{M}(|i\rangle\langle j|)$ its Choi matrix, and $\mathcal{D}_0(X) = \mathrm{Tr}(X)I/2$ the completely depolarizing channel, whose Choi matrix is $I/2$. Suppose $J_\mathcal{M}$ is positive definite and $\mathcal{M} \neq \mathcal{D}_0$. Since $\mathrm{Tr}\, J_\mathcal{M} = 2$, the number $q_\mathcal{M} = 1-2\lambda_{\min}(J_\mathcal{M})$ lies in $(0,1)$. The map $\mathcal{T} = [\mathcal{M} - (1-q_\mathcal{M})\mathcal{D}_0]/q_\mathcal{M}$ is trace preserving and unital, and its Choi matrix $[J_\mathcal{M} - (1-q_\mathcal{M})I/2]/q_\mathcal{M}$ is positive semidefinite, so $\mathcal{T}$ is a unital channel. Because $\mathcal{T} \circ \mathcal{D}_q = q\mathcal{T} + (1-q)\mathcal{D}_0$ for every unital $\mathcal{T}$,

$$\mathcal{M} = \mathcal{T} \circ \mathcal{D}_{q_\mathcal{M}}, \qquad \mathcal{M}^{\otimes n} = \mathcal{T}^{\otimes n} \circ \mathcal{D}_{q_\mathcal{M}}^{\otimes n}. \tag{43}$$

In words, $\mathcal{M}$ hides a fully depolarizing part of weight $1 - q_\mathcal{M} = 2\lambda_{\min}(J_\mathcal{M})$. For a Pauli channel with error probabilities p_I, p_X, p_Y, p_Z, this weight is $4\min_i p_i$.

The memory construction adapts as well. A unital qubit channel acts on Bloch vectors through a real matrix, $\mathcal{M}(\frac{1}{2}(I + v\cdot\sigma)) = \frac{1}{2}(I + \Lambda_\mathcal{M} v\cdot\sigma)$. Let $s_\mathcal{M}$ be the largest singular value of $\Lambda_\mathcal{M}$, positive because $\mathcal{M} \neq \mathcal{D}_0$, and choose unit vectors with $\Lambda_\mathcal{M} a = s_\mathcal{M} b$. The input states with Bloch vectors $\pm a$ leave the channel as the commuting states $\frac{1}{2}(I \pm s_\mathcal{M}\, b\cdot\sigma)$. So store the bit along a and read it along b: conjugate the data block of Eq. (27) by single-qubit rotations, from the b basis before and into the a basis after. Each data qubit is then an independent copy of the stored bit, flipped with probability $(1 - s_\mathcal{M})/2$.

Corollary 8 (Sharp law for anisotropic unital noise)**.** *Let $\mathcal{M} \neq \mathcal{D}_0$ be a unital qubit channel with positive definite Choi matrix, let the noise be $\mathcal{M}^{\otimes n}$, and define $W_{\mathcal{M},k}$ like $W_{q,k}$. For all sufficiently large k and $0 < \varepsilon < s_\mathcal{M}/2$,*

$$\begin{aligned} \tfrac{1}{4}\, e^{\alpha_{s_\mathcal{M}}(k-3)} \log\frac{s_\mathcal{M}}{\varepsilon} &\le W_{\mathcal{M},k}(\varepsilon) \\ &\le \left\lceil A_{q_\mathcal{M}}\, 2^{\beta_{q_\mathcal{M}} k} \log\frac{2}{\varepsilon} \right\rceil. \end{aligned} \tag{44}$$

Hence $W_{\mathcal{M},k} = \exp[\Theta_\mathcal{M}(k)]$ at fixed accuracy.

Proof. The upper bound is Theorem 7 with $\mathcal{R}_t = \mathcal{T}^{\otimes n}$ and $\mathcal{S}_t = \mathrm{id}$. For the lower bound, the rotated reservoir has exactly the structure of Sec. V with q replaced by $s_\mathcal{M}$: the logical bit flips with probability p_r computed from $p = (1 - s_\mathcal{M})/2$, the readout $b\cdot\sigma$ on the first data qubit gives $f_b(t) = (-1)^b s_\mathcal{M} \lambda_r^t$, and Eqs. (29)–(31) hold with $\alpha_{s_\mathcal{M}}$. □

For dephasing-biased Pauli noise with $p_X = p_Y = 0.01$ and $p_Z = 0.08$, the depolarizing core has weight 0.04, so $q_\mathcal{M} = 0.96$. The memory is stored along Z, which only X and Y errors flip, so $s_\mathcal{M} = 0.96$ as well: the law holds with the same exponents as depolarizing noise at $q = 0.96$.

The upper bound sees only the depolarizing core, through $q_\mathcal{M}$; the lower bound sees the best-preserved Bloch direction, through $s_\mathcal{M} \le q_\mathcal{M}$. Both exponents are linear in k, but near the boundary of the set of channels the constants can lie far apart: for $p_X = 0.001$ and $p_Y = p_Z = 0.2$, the core has weight 0.004 although $s_\mathcal{M} \approx 0.6$, and $\beta_{q_\mathcal{M}} \approx 6 \times 10^4$.

C. Cooling noise: memory that grows with the register

Theorem 7 charges nonunital noise as fresh-state supply. Is the charge fair, or an artifact of the proof? Amplitude damping, AD_γ, relaxes $|1\rangle$ to $|0\rangle$ with probability γ and has $\mathrm{AD}_\gamma(I) = \mathrm{diag}(1+\gamma, 1-\gamma)$. A layer $\mathrm{AD}_\gamma^{\otimes n}$ thus supplies up to $n\log_2(1+\gamma)$ fresh qubits' worth of purity per cycle, and a reservoir can indeed spend this supply.

The mechanism is the refrigerator of Ref. [2]. Damping drives idle qubits toward $|0\rangle$, so a block left alone for a few cycles becomes a usable stand-in for freshly prepared qubits. A fixed reversible circuit can keep $s+1$ blocks of r qubits in rotation: one block holds the data while the others cool. In every cycle, the circuit decodes the data block by a threshold vote, writes the decoded bit into the coolest block, and sends the old data block to the back of the cooling queue. Only the two program qubits that select hold or write are reset.

Theorem 9 (Cooling noise)**.** *Fix $0 < q, \gamma < 1$, and let every cycle end with the noise $(\mathcal{D}_q \circ \mathrm{AD}_\gamma)^{\otimes n}$. (i) For every register size, encoding, fixed unitary, and bounded readout,*

$$W_f(\varepsilon) \le \left\lceil A_q\, 2^{\beta_q k} (1+\gamma)^{\beta_q n} \log\frac{2}{\varepsilon} \right\rceil. \tag{45}$$

(ii) With only $k = 2$ fresh qubits per cycle, there is a reservoir with

$$W_f(\varepsilon) \ge \left\lfloor e^{cr}/4 \right\rfloor, \qquad r = \left\lfloor \frac{n-2}{s+1} \right\rfloor, \tag{46}$$

for all $0 < \varepsilon < 1/2$, where $c = c_{q,\gamma} > 0$ and $s = s_{q,\gamma} \in \mathbb{N}$ depend only on q and γ. At any fixed $k \ge 2$ and fixed accuracy, the longest readable memory is therefore $\exp[\Theta_{q,\gamma}(n)]$.

Part (i) is Theorem 7 with $\mathcal{S}_t = \mathrm{AD}_\gamma^{\otimes n}$ and $J_\mathcal{S} = (1+\gamma)^n$. Appendix D proves part (ii) with explicit constants. On computational-basis states, the noise acts as a bit channel with $\Pr(1|0) = a = (1-q)/2$ and $\Pr(1|1) = a + d$, where $d = q(1-\gamma)$; its stationary probability of a 1 is $p_* = a/(1-d) < 1/2$. One can take $c = d^2(1/2 - p_*)^2/2$ and s the least integer with $d^s \le (1/2 - p_*)/2$. Without damping, $p_* = 1/2$, and the pipeline has nothing to cool with, as Theorem 1 demands.

The cooling route is real but expensive. At $q = 0.9$ and $\gamma = 0.3$, a pipeline with $s = 8$ and roughly 1100 qubits reaches a logical error of about 6×10^{-7} per cycle in stationary operation, with two fresh qubits per cycle. The majority reservoir of Sec. V reaches 1.8×10^{-7} with 32 qubits, 17 of them fresh in every cycle [6]. *Resets supply k fresh qubits per cycle, damping supplies $\Theta(n)$, and the memory is exponential in either supply.*

D. The assumptions are needed

Proposition 10 (Two counterexamples)**.** *(i) Under pure dephasing $\mathcal{Z}_p(X) = (1-p)X + pZXZ$ on every qubit, $0 < p < 1/2$, a reservoir with $k = 2$ keeps a written bit perfectly readable forever. (ii) Let every qubit carry a Pauli operator $P_i \in \{I, X, Y, Z\}$, drawn once with probabilities $\frac{1+3q}{4}$, $\frac{1-q}{4}$, $\frac{1-q}{4}$, $\frac{1-q}{4}$ and applied in every cycle. Averaged over the draw, one cycle of this noise is exactly $\mathcal{D}_q^{\otimes n}$, yet the same reservoir reads the written bit perfectly in every second cycle. In both cases, no finite window achieves $W_f(\varepsilon)$ with $\varepsilon < 1$ on histories that contain a write.*

Proof. Let a data qubit lie outside the reset register, which holds program qubits (c, b). The unitary swaps b into the data qubit when $c = 1$ and acts trivially when $c = 0$; the old data value leaves with the program register at the next reset. After write(b) and $t-1$ holds, the data qubit is $\mathcal{Z}_p^t(|b\rangle\langle b|) = |b\rangle\langle b|$ in case (i), since computational-basis states are fixed points, and $P^t|b\rangle\langle b|P^t$ in case (ii), which equals $|b\rangle\langle b|$ for even t because $P^2 = I$. Averaged over the draw, $\mathbb{E}[P\rho P] = \frac{1+3q}{4}\rho + \frac{1-q}{4}(X\rho X + Y\rho Y + Z\rho Z) = \mathcal{D}_q(\rho)$. Now take two histories that differ only in a bit written before the last L inputs, at a time chosen so that an even number of noisy cycles separates the write from time 0. Their Z readouts at time 0 are ± 1, while their last L inputs are the same holds, so no function of these inputs lies within $\varepsilon < 1$ of both outputs. □

Case (i) has a singular Choi matrix and no depolarizing core: dephasing preserves a classical sector. Case (ii) has a depolarizing single-cycle marginal but no depolarizing layer in any realization; the environment stores each error and returns it. Theorem 7 therefore needs the depolarizing layer to act in every cycle of every realization, which is the Markovian setting. Non-Markovian dynamics can likewise extend reservoir memory in practice [7]. In both cases the driven state is not unique, so W_f is read on histories that contain a write.

Any noise is allowed around a layer of independent depolarization: unital noise is free, and nonunital noise counts as fresh qubits.

VIII. DISCUSSION

A. Relation to previous work

Table II separates gate restrictions, fresh-state resources, and the quantity shown to forget, at the theorem level. The tools of Sec. IV are established inequalities [3, 4]; we do not claim new hypercontractivity. The majority mechanism is also standard error-correcting logic [8]. The result is their explicit combination into a uniform arbitrary-dynamics obstruction, with matching reset-width scaling and a task-learning consequence.

Our results add a reset axis to the classification of Ref. [2]. Without fresh ancillas, depolarizing-type noise allows only logarithmic time, which matches the $O(\log n)$ washout of an unrefreshed register in Sec. IV E, while amplitude-damping-type noise allows exponential time. With k fresh qubits per cycle, depolarization gives $\exp[\Theta(k)]$ (Theorem 1), and damping on top gives $\exp[\Theta(n)]$ already at constant k (Theorem 9). The cooling mechanism behind the last lower bound is that of Ref. [2], which also underlies measurement-free error correction under nonunital noise [10]. What Theorems 7 and 9 add is the matching upper bound and a single fresh-state accounting that treats resets and nonunital noise alike.

The lower construction does not contradict the Clifford result. The majority-controlled update is not generally Clifford. The construction uses reversible nonlinear Boolean processing, while its states and outputs remain classically describable. Thus no entanglement or classical simulation hardness is required to attain the memory-resource scaling.

The feature expansion is not the novelty. Fading-memory Volterra approximation is classical [12], and quantum temporal Fourier representations are established [5]. An expansion does not by itself bound the required window from local physics. Here the window is physically derived, its dependence on reset width is unavoidable, the coefficients are uniformly bounded, and the learner does not need them in advance.

B. Scope and interpretation

The upper bound needs a fixed-strength layer of independent depolarization on every qubit in every cycle of every noise realization; any noise may surround it, with nonunitality counted as fresh-state supply (Sec. VII). Proposition 10 shows that neither the layer nor Markovianity can be dropped. The theorem permits general averaged CPTP updates only when their nonunitality satisfies Eq. (8); it does not include a persistent noiseless external controller, and it does not assume a local or efficiently simulable U. The matching lower bounds use the same ideal-unitary/noise ordering and product noise; they are not fault-tolerance theorems with noise inserted after every elementary gate of a majority circuit.

All upper guarantees are uniform over inputs; independence is needed only for the supervised-learning statement. Correlated overlapping windows and closed-loop forecasting require separate statistical or stability arguments. The constants in Eq. (21) grow rapidly as $q \uparrow 1$, and for anisotropic noise the upper constant sees only the depolarizing core: we prove the order in k and n at fixed noise, not optimal noise-dependent exponents or a practical hardware crossover. Likewise, the fresh-state accounting is an upper-bound currency: a channel that replaces every qubit by a fixed state carries a large $J_{\mathcal{R}}$ but no memory at all. The library (38) is a worst-case cer-

TABLE II. Comparison with related results. The assumptions do not form a single hierarchy. In particular, bounded resets with arbitrary gates neither subsume nor follow from unrestricted resets with Clifford gates. All comparisons are restricted to the stated noise and update models.

Result	Dynamical resources	Relevant conclusion and distinction
Aharonov *et al.* [9]	No continually supplied fresh inputs; noisy reversible or quantum computation	An exponential size cost for useful depth. Does not give the bounded-reset driven-memory law.
Ben-Or *et al.* [2]	No fresh ancillas; general local noise	Computation for logarithmic (depolarizing-type), polynomial (dephasing-type), or exponential (amplitude-damping-type) time. No bounded-reset converse.
Shtanko and Sharma [10]	Geometrically local circuits; weak nonunital noise; no mid-circuit measurements	Error correction and arbitrary depth with polylogarithmic overhead.
Nelson *et al.* [8]	Clifford gates, arbitrary reset placement, local depolarization	Trace-distance forgetting in $O(\gamma^{-1}\log n\log(n/\varepsilon))$ layers, $\gamma = 1-q$. Allows more resets but fewer gates.
Mele *et al.* [11]	Random circuits; general local nonunital noise	Effective-depth bounds for observable expectation values in an ensemble setting, rather than a common contraction for every driven update.
Sannia *et al.* [7]	Quantum reservoirs with non-Markovian dynamics	Memory beyond Markovian exponential fading.
This work	Arbitrary gates; k resets per cycle; Markovian noise with a depolarizing layer	$\exp[\Theta(k)]$ driven-window scaling for unital noise; $\exp[\Theta(n)]$ at constant k under damping; nonunital noise priced as fresh-state supply; task-label learning.

tificate, not necessarily the best classical model; and, as the one-scalar majority example shows, a window-length lower bound is not a classical complexity lower bound.

For fixed noise, fixed k, and bounded readout, the known library and convex learner exclude an exponential asymptotic separation in supervised task learning when the classical learner has polynomially larger resources and the same task-label access. This does not simulate the quantum state, sample its full output distribution, or identify a circuit's output from its description without data. It also does not promise the same risk at the same finite training budget.

C. Conclusion

The number of qubits and the supply of fresh qubits control different aspects of noisy memory. Register size can delay the loss of an arbitrary initialization. Fresh-state supply, whether it comes from resets or from nonunital noise, controls how long newly written information can persist during driven operation. We give a uniform converse, a matching local-readout construction, their extension to any unital noise around a depolarizing core and to every nontrivial unital qubit channel with a full-rank Choi matrix, a tight cooling regime in which the memory grows exponentially with the register, and the resulting classical-learning guarantee. Within the stated noisy-update model, a readable lifetime T costs $\Theta(\log T)$ fresh qubits per cycle. *Qubits hold a memory; fresh qubits keep it.*

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DATA AVAILABILITY

No data were created or analyzed in this study.

Appendix A: Proof of Lemma 3

Write $Y = Y_+ - Y_-$ with $Y_\pm \geq 0$ of orthogonal supports, and put $G_\pm = 2Y_\pm/s$. Since Y is traceless, $\tau G_\pm = 1$, and $|G_\pm|_2 \leq 2t/s$. For positive G with $\tau G = 1$, scalar Hölder interpolation on its eigenvalues gives

$$1 \leq (\tau G^{\bar{p}})^{1/(2-\bar{p})}(\tau G^2)^{(1-\bar{p})/(2-\bar{p})}, \tag{A1}$$

hence $|G|_{\bar{p}} \geq |G|_2^{-a}$ with $\bar{p} = 1-q$. For any measurement effect $0 \leq M \leq I$, either $\tau M \geq 1/2$ or $\tau(I-M) \geq 1/2$. Since $M^{\bar{p}} \geq M$, the corresponding $\bar{p}$-quantity is at least c_q. Applying Eq. (14) to the appropriate error term therefore gives

$$\begin{aligned} e(M) &:= \tau[(I-M)\mathcal{N}_q G_+] + \tau[M\mathcal{N}_q G_-] \\ &\geq c_q\Big(\frac{s}{2t}\Big)^a. \end{aligned} \tag{A2}$$

Here $e(M)$ is the total error of the test M for telling the noisy positive part from the noisy negative part. The variational formula for the trace distance is $|\mathcal{N}_q Y|_1 = s[1 - \min_M e(M)]$, which proves Eq. (17). □

Appendix B: The initialization term is necessary

Take $U = I$ and let the reset register be disjoint from $m = n - k$ data qubits. Initialize the latter in $|0\rangle^{\otimes m}$ or $|1\rangle^{\otimes m}$. After L cycles, their independent Z outcomes have mean $\pm q^L$. Their sum has mean $\pm m q^L$ and variance at most m. Chebyshev's inequality bounds the majority-decoding error under either initialization by $1/(mq^{2L})$. Consequently,

$$\left\|\rho_L^{(0)} - \rho_L^{(1)}\right\|_1 \geq 2\Big(1 - \frac{2}{mq^{2L}}\Big). \tag{B1}$$

For $L = c \log m$ with $c < 1/(2|\log q|)$, the right side tends to two. Thus an $\Omega_q(\log n)$ initialization term is unavoidable at fixed k. This is an initialization result, distinct from the driven lower bound in Sec. V.

Appendix C: Learning: proof, solver, and precision

Proof of Theorem 6. Let $\mathcal{G} = \{\mathrm{clip}_{[-1,1]}(c^\top \psi) : c \in [-1,1]^K\}$. We first bound the generalization gap uniformly over $\mathcal{G}$ using a finite grid, solely for the proof. Round each coefficient to a grid in $[-1,1]$ of spacing $1/(KN)$. There are $(2KN+1)^K$ grid points, and rounding changes every prediction by at most $1/N$. Clipping is 1-Lipschitz; squared loss on $[-1,1]$ is 4-Lipschitz and lies in $[0,4]$. Hoeffding's inequality [13] and a union bound give, for the empirical risk $\hat{\mathcal{R}}$,

$$\begin{aligned}\sup_{g\in\mathcal{G}} \big|\mathcal{R}(g) - \hat{\mathcal{R}}(g)\big| &\leq D_N \\ &:= \frac{8}{N} + \sqrt{\frac{8}{N}\Big(K\log(2KN+1) + \log\frac{2}{\zeta}\Big)}.\end{aligned} \tag{C1}$$

The $8/N$ accounts for rounding both population and empirical losses. By Eq. (37), the physical window output f_L has coefficients in the box and takes values in $[-1,1]$. Clipping never increases squared loss against a label in that interval. Consequently, $\hat{\mathcal{R}}(\hat{f}) \leq F_N(\hat{c}) \leq \hat{\mathcal{R}}(f_L) + \nu$. Apply Eq. (C1) twice. Finally, $|f_L - f| \leq \varepsilon$ gives $\mathcal{R}(f_L) \leq \mathcal{R}(f) + 4\varepsilon$. Thus $\mathcal{R}(\hat{f}) \leq \mathcal{R}(f) + 4\varepsilon + \nu + 2D_N$, which is Eq. (41). The grid is not enumerated by the algorithm; it replaces a coarse complexity bound with an elementary covering argument. □

A polynomial-time optimization witness. Feature evaluation costs $O(KL)$ arithmetic operations per example, in addition to evaluating the input coordinates. Since $\nabla^2 F_N = \frac{2}{N}\sum_i \psi(H_i)\psi(H_i)^\top \preceq 2KI$, projected gradient descent, $c^{(0)} = 0$ and $c^{(s+1)} = \mathrm{proj}_{[-1,1]^K}[c^{(s)} - \nabla F_N(c^{(s)})/(2K)]$, provides a simple explicit solver. For any minimizer c_*, the standard smooth-convex projection inequality gives

$$\begin{aligned}&F_N(c^{(s+1)}) - F_N(c_*) \\ &\qquad \leq K\big(\|c^{(s)} - c_*\|_2^2 - \|c^{(s+1)} - c_*\|_2^2\big).\end{aligned} \tag{C2}$$

It follows directly by minimizing the quadratic upper bound to F_N over the box, using its $2K$-strongly convex quadratic term, and then using convexity at c_*. Summing and averaging, with $\|c_*\|_2^2 \leq K$, shows that the averaged iterate $\bar{c}_S = \frac{1}{S}\sum_{s=1}^S c^{(s)}$ satisfies $F_N(\bar{c}_S) - F_N(c_*) \leq K^2/S$. Thus $S = \lceil K^2/\nu \rceil$ iterations suffice when $\nu > 0$, each costing $O(NK)$. This conservative witness requires no covariance-conditioning assumption.

Finite precision does not require quantum simulation. Suppose each encoding coordinate is evaluated within ξ and clipped to $[-1,1]$. By telescoping, a product of L such coordinates changes by at most $L\xi$, and a prediction with coefficients in the box by at most $KL\xi$. Both unclipped predictions lie in $[-K,K]$, so each training loss changes by at most $2(K+1)KL\xi$. For a target optimization tolerance $\nu > 0$, take $\xi \leq \nu/[8(K+1)KL]$: the approximate objective is then uniformly within $\nu/4$ of the exact one, so solving it to tolerance $\nu/2$ achieves exact empirical suboptimality at most ν, and the feature error at prediction is bounded in the same way. For the quadratic objective, the exact gradient step with step size $1/(2K)$ is nonexpansive, since its linear part has eigenvalues in $[0,1]$, and so is projection onto the box. Per-iteration Euclidean rounding error ω therefore accumulates to at most $S\omega$ after S iterations; since the objective is $2(K+1)\sqrt{K}$-Lipschitz on the box, ω inverse-polynomial in S, K, and $1/\nu$ controls the remaining optimization error. All necessary bit lengths and iteration counts are polynomial, and no knowledge of the unknown quantum-dependent coefficients is used.

Appendix D: Proof of Theorem 9(ii)

Bit channel. Amplitude damping maps $|0\rangle\langle 0|$ to itself and $|1\rangle\langle 1|$ to $(1-\gamma)|1\rangle\langle 1| + \gamma|0\rangle\langle 0|$; depolarization then flips each bit with probability $(1-q)/2$. On diagonal states, one cycle of noise is therefore the bit channel $\Pr(1|x) = a + dx$ with $a = (1-q)/2$ and $d = q(1-\gamma) \in (0,1)$, whose stationary probability of a 1 is $p_* = a/(1-d) = (1-q)/[2(1-q+q\gamma)] < 1/2$. All states below stay diagonal, so every data qubit is a classical bit that evolves independently under this channel. After s steps from the value x, it is 1 with probability $p_* + d^s(x - p_*) \leq p_* + d^s$. Let s be the least integer with $d^s \leq (1/2 - p_*)/2$; then this probability is at most $u = (1/2 + p_*)/2 < 1/2$.

Reservoir. The reset register holds two program qubits (c,b), prepared in $|0,0\rangle$ for hold and in $|1,b\rangle$ for write(b). The other qubits form $s+1$ blocks $x_0, \ldots, x_s \in \{0,1\}^r$; spare qubits stay idle, so $r = \lfloor (n-2)/(s+1) \rfloor$. With the threshold $\vartheta = a + d/2$, let $F_{c,b}(x) = \mathbf{1}\{|x| \geq \vartheta r\}$

for $c = 0$ and $F_{c,b}(x) = b$ for $c = 1$, where $|x|$ counts ones. The fixed unitary is the basis permutation

$$U : |c, b\rangle|x_0, x_1, \ldots, x_s\rangle \longmapsto |c, b\rangle|x_s \oplus F_{c,b}(x_0)1_r,\, x_0, \ldots, x_{s-1}\rangle. \tag{D1}$$

It is reversible: the output reveals $x_0, \ldots, x_{s-1}$, hence $F_{c,b}(x_0)$, hence x_s. Block x_0 holds the data; a block leaving it passes through positions $1, \ldots, s$ and is reused after s noisy cycles.

One cycle. Suppose the data block written at some cycle carries the bit F; for a write, $F = b$ exactly. The reused block entered the queue s noisy cycles earlier, so given the past its bits are independent and each is 1 with probability at most u. After the XOR, each bit of the new data block differs from F with probability at most u, independently. After this cycle's noise, each bit is 1 with probability at least $a + d(1 - u)$ if $F = 1$ and at most $a + du$ if $F = 0$, again independently. The threshold ϑr lies a margin $d(1/2 - u) = d(1/2 - p_*)/2$ away from both means, so Hoeffding's inequality [13] bounds the probability that the next cycle decodes $1 - F$ by

$$e^{-2r[d(1/2-u)]^2} = e^{-cr}, \qquad c = \frac{d^2(1/2 - p_*)^2}{2}. \tag{D2}$$

Window bound. Consider two driven histories that agree before time $-L$, carry write(0) or write(1) at time $-L$, and then receive L holds; the driven state exists by part (i). By a union bound, all L decodings and a final threshold decision on the data block after the noise at time 0 are correct with probability at least $1-(L+1)e^{-cr}$. The diagonal observable $O = 2\,\mathbf{1}\{|x_0| \geq \vartheta r\} - 1$ has $\|O\|_\infty = 1$, and its expectations in the two histories differ by at least $2-4(L+1)e^{-cr}$, which is at least 1 whenever $(L + 1)e^{-cr} \leq 1/4$. Any function of the last L inputs, which are holds in both histories, then errs by at least $1/2 > \varepsilon$ on one of them. Hence $W_f(\varepsilon) \geq \lfloor e^{cr}/4 \rfloor$ for all $\varepsilon < 1/2$. Since $r \geq (n - 2)/(s + 1) - 1$, the memory is $\exp[\Omega_{q,\gamma}(n)]$ at constant k. □

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